From scientific description to independent computational reconstruction

Paper II: a prospectively frozen reconstruction study linked to the EPJ commercial-software corpus (main study sealed; human validation pending)

DRAFT — NOT SUBMISSION READY: main study sealed; delegated mechanical adjudication; A/B/C/E AI-assisted review imported as a provisional sensitivity layer; investigator sign-off, human adjudication and human validation (Section D) pending

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Abstract (draft)

Access to research artifacts does not establish whether the published scientific description supports an independent implementation. This prospectively frozen study estimated publication-grounded computational reconstructability under a specified agent, information boundary and resource budget, without consulting the original implementation or purchasing study-specific commercial software. The source audit identified 10,000 deposited records representing 9,935 distinct PMIDs; the unique-PMID frame was approved while every deposited row was preserved. Within ACCESS, resource availability was distinguished from input-state identifiability and historical-state retrievability. Eligibility, input and resource access, specification, implementation, execution, numerical agreement and target-linked conclusion preservation were assessed as separate levels. Three fixed independent blind run slots per paper supported majority, strict and permissive outcome definitions while missing or contaminated slots were retained as unresolved. Targets, tolerances and resource ceilings were frozen before every run. After a 7-paper prospective pilot and protocol freeze, 100 papers were sampled from 461 route-eligible papers in 7 strata. Only 10 reached blinded independent reconstruction under the prespecified access, input and specification rules; 90 stopped before reconstruction with machine-readable barrier states. Delegated mechanical adjudication of the 30 sealed slots found no paper meeting the majority criterion; a separate AI-assisted verification review, provisional and not investigator-signed, proposes 1 and does not overwrite the sealed result. Verification of the 90 non-executed papers supported most recorded barriers and identified a small number of pre-execution misclassifications from an overly conservative target-leakage detector. Large pre-reconstruction attrition is the primary empirical finding; the reconstruction result is conditional on attempt, bounded by the resource ceiling, and does not show human impossibility. Human validation (Section D) has not been performed and remains pending.

Keywords: computational reconstruction; reproducibility; scientific specification; research infrastructure; software accessibility; research agents

1 Introduction

Computational reproducibility terminology varies across communities [1, 2]. ACM's post-2020 terminology distinguishes execution of authors' artifacts from results obtained using independently

developed artifacts [3]. We therefore use publication-grounded independent computational reconstruction as an operational description rather than proposing a universal definition.

The EPJ antecedent examines commercial software and version accessibility [4]. Those conditions motivate, but do not answer, whether the scientific description contains enough operational detail to implement the reported method independently. The Wet antecedent concerns environmental confounding in clonal systems [5]. Its extension here is a conceptual analogy: nominal materials or data may be available while operational specification remains incomplete. The Wet full text was not accessible in the preparation audit; no detailed empirical claim is imported from it.

Independent paper-to-code reconstruction already has substantial precedent. PaperBench restricts original implementations but supplies author clarifications [6]; ReplicationBench uses curated astrophysics tasks and supplied execution information [7]. Paper2Code studies repository generation [8], while ScienceAgentBench evaluates curated scientific programming tasks [9]. CORE-Bench instead supplies original code and data [10]. Their endpoints and information packages differ and must not be pooled as a publication-level reconstructability rate.

The contribution is a finite-corpus estimation study with an explicit publication-only information boundary and zero study-specific software expenditure. It makes no first-study or terminological novelty claim. Figure 1 shows the conceptual framework and the exclusion of robustness experiments.

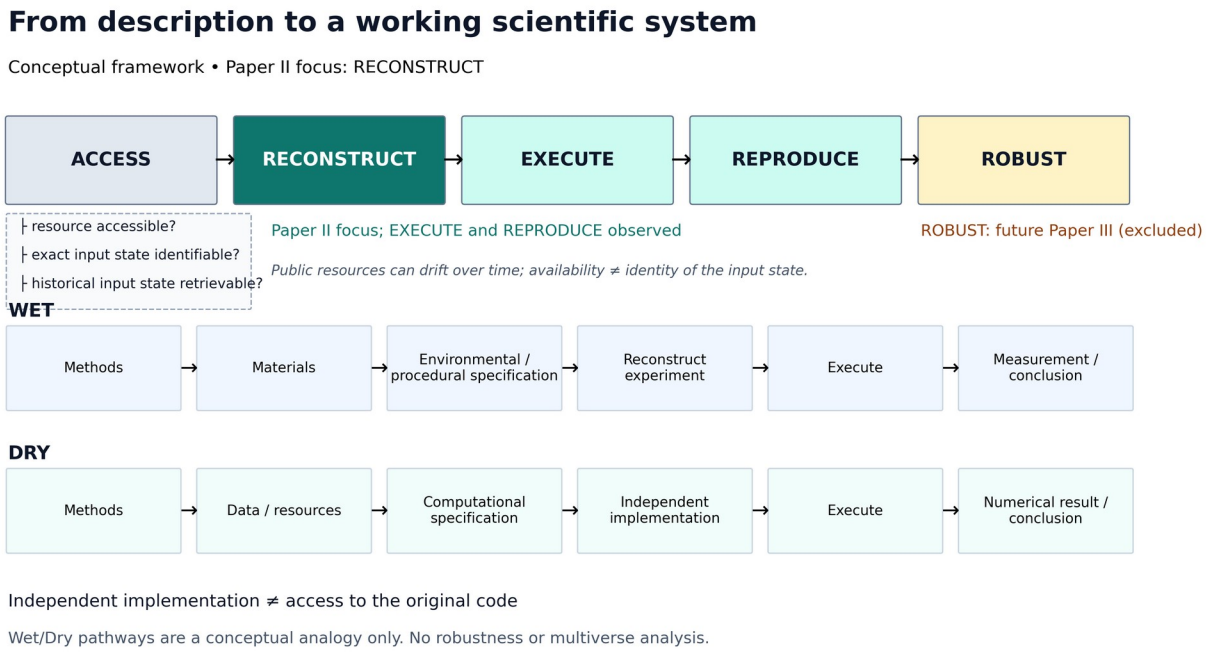


Figure 1. Conceptual Wet/Dry framework. ACCESS is refined internally into three questions (resource accessible; exact input state identifiable; historical input state retrievable) because public resources can drift over time. Paper II focuses on RECONSTRUCT and observes EXECUTE and REPRODUCE. ROBUST belongs to future Paper III and is excluded. The two pathways are an analogy, not a literal equivalence or an empirical result.

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2 Methods (procedures as executed under the prospectively frozen protocol; amendments in the supplement)

2.1 Source frame and eligibility

Deposited source records were preserved and paper identity was derived separately. G1 assessed whether a central claim depends on a computational procedure with a potentially objective output; G2 identified the principal target; G3/G4 assessed lawful input and zero-purchase resource access; G5 described specification sufficiency. Under the amended candidate criterion (AMEND-2026-09-24-03) primary reconstruction required deposited input data that a third party can legally and immediately obtain without contacting the original authors. Exhaustive route screening of the unique-PMID frame identified 461 route-eligible papers; every screening failure, including the 7 ACCESSIBILITY_GATE_FAILED papers, was retained as an auditable attrition layer. Neither access nor specification failures excluded otherwise eligible papers from the all-eligible policy estimand. Historical code/data flags were not used in place of these assessments. G1–G5 were recorded as delegated mechanical adjudications; investigator verification is pending.

2.1a Input accessibility versus input-state identifiability

Input accessibility was distinguished from input-state identifiability. Within ACCESS three questions were kept separate: whether a named resource is accessible; whether the exact state of the input data used in the original analysis can be uniquely identified from the publication and associated records (input-state identifiability); and whether that exact previously used state can still be obtained by an independent third party (historical-state retrievability). For externally maintained public datasets that authors could not necessarily redistribute or freeze, we recorded, where available, the repository or accession, release or version, retrieval date or timestamp, query parameters, file identity, schema or release identifier, and cryptographic checksum. Public availability alone was not treated as evidence that the exact historical analytical input remained identifiable or retrievable. This refinement is descriptive: the frozen G1–G5 gates and barrier states were not rescored, and no dataset drift was measured. Table S1 maps these reporting requirements to framework stages; Table S2 summarizes the fields actually recorded in the frozen main-study records.

2.2 Sampling and target selection

A 7-paper prospective pilot (one paper per field, three blind slots each, 21 slots) was executed under prospectively frozen specifications and used only to identify procedural defects; its prespecified review recorded one defect (PD-01, an internally inconsistent token ceiling) and did not change target-selection, tolerance, conclusion or taxonomy rules, and no pilot success rate was computed. The final protocol and SAP were then frozen with hashes and RFC 3161 timestamps (PROTOCOL-FREEZE-2026-09-25). From the 461 route-eligible papers, 100 were sampled by stratified random sampling using a deterministic disjoint assignment of the 7 overlapping EPJ fields; within-field sampling probabilities and weights were saved. Before any run, one central target per paper was selected by the frozen hierarchy and its associated claim, input identifiers, allowed and blocked resources, metric, numerical and conclusion criteria, resource ceiling and stopping rule were frozen. No outcome-informed target replacement occurred. Planning precision across candidate sample sizes is reported in Supplementary Table S11; it is analytic, not empirical evidence.

2.3 Independent reconstruction intervention

Each attempted paper received three clean, fixed blind slots with the same target, input package and permissions and equivalent model and resource settings (per slot: 40 tool calls, 1.5 million

provider tokens, 3,600 s per step, 14,400 s wall, one CPU, 3 GB, no network). Permitted materials were the publication, methodological supplement, vetted deposited inputs and general documentation; original implementations and other runs were blocked. A functioning reconstruction harness was implemented and used: a custodian/broker served only manifest-listed items to an isolated worker, enforced resource stops and denied author implementation material and reveal-class items to the solver, with sealed event chains and per-slot isolation. These controls passed synthetic qualification probes; model-side retrieval-tool restriction for provider-hosted tools and cross-slot budget comparability were not formally qualified, and complete exclusion of prior model-training exposure cannot be certified. Papers that failed a frozen pre-execution rule were recorded as non-executed with a machine-readable state and were not run.

2.4 Outcomes and comparison

L1 specification, L2 implementation, L3 execution, L4 numerical reproduction and L5 conclusion preservation were scored separately. A run succeeded only when an independent implementation executed and met the frozen target-specific numerical rule. Majority paper success required two clean successes among three fixed slots; strict and permissive definitions were prespecified sensitivities. Missing or contaminated slots remained unresolved, with bounds rather than reduced denominators. Exact counts, displayed-precision scalar comparisons and target-specific stochastic rules replaced arbitrary universal tolerances. Matching numbers do not by themselves prove implementation fidelity.

2.5 Statistical analysis and failure attribution

Design-weighted finite-corpus estimates were computed with separately reported sampling uncertainty and unknown-status identification bounds; fixed-frame census counts carry no paper-sampling error. Complete-triple consistency, hierarchical outcomes, failure codes and burden were reported. No multivariable model was fitted and code-sharing associations were not interpreted causally. Observed reconstruction failure was distinguished from demonstrated publication inadequacy; the resource ceiling has its own failure code. Details and prespecified sensitivities are in the frozen SAP.

2.6 Freeze, reveal and human validation

Targets were sealed before runs and all outcomes were sealed before any original-code reveal. Delegated mechanical adjudication of the sealed slots followed the frozen tolerance rules (AMEND-2026-09-25-07). The reveal was descriptive only: no original code was executed and no blind score was revised. A separate AI-assisted evidence review of the sealed slots (A), the non-executed states (B), the historical G1–G5 and pilot records (C) and the reveal (E) was imported as a provisional verification-adjusted sensitivity layer that awaits investigator sign-off. Human validation (Section D) is prespecified as outcome-balanced human cases scored by validators masked to agent and reveal results; the frozen records support 10 cases (one per attempted paper). It has not been performed: no validator has been recruited and no institutional determination obtained. No human outcome is simulated. No robustness or multiverse analysis is part of Paper II.

3 Source corpus audit

The two pinned source CSVs contain 10,000 rows and 9,935 unique PMIDs, with 65 excess records over a unique-paper count. There are 64 repeated identifiers: 63 occur twice and 1 occurs three times. The deposit has 9,996 distinct PMID-field memberships, including 4 within-field excess records and 60 PMIDs spanning fields. Non-field extracted values agree across repeated records. Table 1 presents source-row and within-field identity counts; it is not an eligibility or reconstruction-results table.

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Table 1. Observed deposited corpus structure.

Recorded EPJ field	Source rows	Unique PMIDs within field
Biomedical Basic	1429	1429
Clinical Medicine	1429	1429
Chemistry Materials	1429	1429
Physics Engineering	1429	1428
Social Behavioral	1429	1429
Computational Science	1429	1429
Environmental Earth	1426	1423

The original sampling code caps annual candidate retrieval and its nominal random-offset expression is always zero. Historical candidate/API snapshots and validation annotations were not recovered. Consequently original population-wide selection probabilities and extraction accuracy are not established by this audit. Main-study outcomes below are delegated mechanical adjudications that await human adjudication; no human-validated reconstruction rate exists.

4 Main-study execution and sealed delegated mechanical results (frozen)

Status: every number in this section is a delegated mechanical adjudication of sealed blind outcomes under frozen rules (AMEND-2026-09-25-07). Human adjudication and investigator verification are pending, and the 7 prospective pilot papers and 7 ACCESSIBILITY_GATE_FAILED papers are outside every denominator below.

Exhaustive route screening of the unique-PMID frame left 461 route-eligible papers, from which 100 were sampled in 7 frozen strata. 10 sampled papers completed three blind slots (30 runs); 90 were not run and retain machine-readable states (Table S3): external public reference resource required, delegated G1–G5 negative, required inputs absent from the public listing, specification rejected by the quote/leakage validator, target value not located verbatim, every candidate input leaking the target, or no input retained.

Among attempted papers the majority endpoint was met by 0 of 10 (secondary conditional rate; Wilson reference 0.00–0.28). Prespecified thresholds — strict: failure 10; majority: failure 10; permissive: failure 9, success 1. Run-level L1–L5 frequencies are in Table S4. Stop reasons: wall_limit 1, tool_limit 16, final_report 10, token_reserve_limit 2, malformed_action_limit 1. Failure codes by run (Table S5): F01 5, F03 2, F13 19, F15 3, F20 2. F13 denotes the frozen resource ceiling, not demonstrated publication inadequacy.

Under the frozen SAP the headline policy estimand retains non-attempted papers as policy non-success. Taking the delegated states as recorded, the simultaneous hypergeometric upper bound on the eligible-population success rate is 0.262 (weighted point estimate 0.000). Because gate verification is pending, an identification envelope that treats every non-attempted paper as unresolved reaches 0.974; it is a bound, not an estimate. No random-intercept model, perturbation, multiverse or alternative-implementation analysis was performed.

Descriptive original-code reveal (Table S6) was recorded only after the seal: 3 attempted papers named public code routes or shipped scripts inside the data deposit (withheld from the solver) and are described; 7 named none and receive a missing reveal assessment. Nothing was executed and no blind score was revisited.

5 Verification-adjusted sensitivity layer (A/B/C/E; provisional, separate from Section 4; D excluded)

Section 4 is frozen. Section 5 is a separate provisional layer pending investigator approval; it does not overwrite Section 4, no reclassified non-executed paper was executed, and the reveal never changed a blind score.

Status: this section reports a separate verification-adjusted sensitivity layer built from an AI-assisted independent evidence review of sections A (30 sealed slots), B (90 non-executed papers), C (historical G1–G5, pilot and access-barrier layers) and E (descriptive reveal). The review is provisional and awaits investigator approval; it is not human-participant validation, no investigator sign-off field was populated, and the sealed blind outcome is byte-identical (SHA-256 350a0e7dcedbb74c…). Every reviewed paper/slot identifier validated against the frozen records and 698 cited evidence references resolved (0 unresolved).

Three estimands are kept separate. Attempt reachability among the 100 selected papers was 10/100 (10.0%; Wilson 95% CI 5.5–17.4%) under both layers. Conditional majority success among attempted papers was 0/10 (0.0%; Wilson 95% CI 0.0–27.8%) under the frozen mechanical adjudication and 1/10 (10.0%; Wilson 95% CI 1.8–40.4%) under the verification-adjusted layer, in which 3 of 30 slot classifications were proposed for change and 27 were supported (PMID:41563087 1/3 → 2/3; Table S7). The observed end-to-end yield under the frozen workflow was 0/100 (0.0%; Wilson 95% CI 0.0–3.7%) frozen and 1/100 (1.0%; Wilson 95% CI 0.2–5.4%) adjusted. None of these is an intrinsic probability that a publication is scientifically reproducible.

Verification of the 90 non-executed papers confirmed 81 recorded states, proposed 7 state corrections and left 2 unresolved (Table S8). Non-confirmed states by frozen category: all_inputs_leak_target 9. Of 10 all_inputs_leak_target flags, 1 confirmed, 7 corrected and 2 unresolved. Whole-token matches inside compressed or binary deposited content and raw input matrices produced false-positive target-leakage flags; this is reported as a methodological finding. Under the adjusted layer the cohort comprises 10 attempted papers, 83 verified pre-reconstruction barriers, 5 papers reclassified as specifiable and runnable but never executed, and 2 unknown. Reclassified papers were not run; the correction is attrition interpretation only, and non-executed papers are not reconstruction failures.

For the eligible-population policy estimand, treating reclassified and unknown papers as unresolved gives a simultaneous sampling-and-identification envelope of 0.002–0.375 (weighted point range 0.010–0.080); treating every non-confirmed leakage classification as unresolved widens the upper bound to 0.401. These are bounds, not estimates.

Section C reviewed 25 historical G1–G5, pilot and ACCESSIBILITY_GATE_FAILED records and supported 25; no post-outcome change was made, the pilot stays outside the main denominator and the access-barrier layer remains auxiliary. Section E reviewed 130 reveal dimension rows for 10 attempted papers (3 with a code route) and proposed 5 conservative completeness corrections, chiefly where deposited downstream scripts start from pre-computed inputs and cannot establish upstream normalisation, filtering or exclusion completeness (Table S9; explicit 15 → 10 dimensions). The reveal remains descriptive; no original code was executed and no blind score was revisited.

Tables S1–S10 are supplied separately in the editable tables file and the supplement.

6 Human validation (Section D) — pending

No human validator has been recruited, no institutional ethics determination has been obtained and no human case has been scored. The Results text, table and figure for Section D are intentionally empty (Table S10). The AI-assisted review in Section 5 is not a substitute for human validation.

7 Discussion boundary and submission hold

The primary empirical message is the large pre-reconstruction attrition: 90 of 100 sampled papers stopped at access, input or specification barriers before any blinded reconstruction under the frozen rules, and verification supported most of these barriers. The 10-paper reconstruction result is conditional on attempt, obtained under frozen resource ceilings and a fixed agent, and is reported side by side as a frozen mechanical value and a provisional verification-adjusted value. Non-executed papers are pre-reconstruction non-executions, not reconstruction failures. Verification identified a small number of pre-execution classification problems, chiefly leakage false positives from whole-token matching in compressed or binary content; the detector's conservatism is a methodological limitation and finding. Agent failure does not prove human impossibility. Original-code availability may correlate with reporting practice but cannot be interpreted causally here. Data/code availability requirements address only selected stages of the reproducibility chain; reconstructability also depends on explicit specification, stable input identity, executable environments and target-linked evaluation rules. This framework is conceptual and is not presented as an empirically validated universal model.

7.1 Dynamic public datasets and input-state identifiability

Public availability is not equivalent to reproducible input identity. Externally maintained public datasets, registries, genomic and administrative databases and APIs may be updated, corrected, reannotated or restructured, and query or API responses may change over time, while retaining the same repository, accession, endpoint or dataset name [11–14]. The same repository, accession or URL therefore does not necessarily denote the same version, and the same version label does not by itself establish the same bytes. This does not imply that all same-accession resources change; it means that public availability does not by itself establish that the exact historical analytical input remains identifiable or retrievable. Three abilities are distinct: the ability to redistribute an input, which authors may lack for third-party data under licence or legal constraints; the ability to identify the exact state that was used; and the ability to retrieve that historical state later. A content hash verifies byte-level identity when the bytes are available [15] but does not recover historical bytes that are no longer archived. Where direct archival redistribution is not permitted, reproducibility must rely on sufficient state identification and provenance: repository or accession, release or version, retrieval timestamp, query parameters, file identity, checksum and transformation provenance [16–20].

Software-version drift, the concern of the EPJ antecedent [4], and dataset-version drift are analogous but distinct: both can separate a nominally available resource from the state actually used. Versioned software may in some settings be preserved or rebuilt more readily than third-party historical data states, although licensing and legacy-version access can also prevent software redistribution. Paper II did not quantify dataset drift and makes no empirical claim about its frequency. The frozen records show only that repository or accession identity was recorded for every sampled paper, that the delegated specifier named external public reference resources for 51 of the 100 sampled papers (119 entries, of which 19 across 11 papers lacked a version in the publication text), that 28 papers stopped at the frozen barrier requiring an unserved external public reference resource (a state that the frozen protocol defined and this refinement leaves unchanged), and that the served inputs were logged with retrieval timestamps and SHA-256 digests but without

repository version identifiers (Table S2). Reference resource identity can matter for results [21]. Input-state identifiability is therefore presented as a conceptual implication, a reporting requirement and a limitation relevant to reconstructability, not as a measured outcome. A focused bibliographic search found no established use of the terms input-state identifiability or historical-state retrievability; they are used provisionally and no novelty is claimed. Data-state reproducibility was not adopted because no established compatible usage was found.

Proposed minimum reporting set for third-party or public input data (a recommendation, not an empirically validated standard), where available: (1) repository or accession; (2) release/version; (3) retrieval date/time; (4) exact query or API parameters; (5) file names/list; (6) schema/version; (7) cryptographic checksum; (8) transformation log; (9) license/redistribution status. Table S1 assigns each element to the framework stage it serves.

Submission is on hold: investigator verification and sign-off of the A/B/C/E proposals, human adjudication of the 30 sealed slots and human validation (Section D) are incomplete. This document must not be submitted as a completed empirical study.

Statements and declarations — pending author completion

Funding, competing interests, author contributions, affiliations and correspondence: not provided. Human-validation ethics/consent determination: not completed. Data/code: the public EPJ-linked repository contains the source audit, frozen protocols, sealed outcome records, adjudication and verification layers and the analysis pipeline. Deposited third-party inputs and full text are retained internally under their applicable rights with hashes and retrieval receipts and are not redistributed.

AI assistance: Devin performed source, literature and design audits, generated the pipeline and draft materials, recorded the delegated G1–G5 and mechanical adjudications, and produced the AI-assisted A/B/C/E review; the blind slots were executed by a separate API-hosted model inside the isolated harness. Human investigators must validate the design, factual claims, adjudications, interpretation and final manuscript, and retain scientific responsibility.

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Antecedent author names follow the supplied author specification. Publisher/Crossref metadata reverse the first author's name components; verify the final bibliographic form before submission. Preprints and version-specific claims are identified in the reference ledger.

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